

Resonant Pre-calculation Theory

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The Question That Breaks Physics

"If time is nothing but entropy, then why do you not see the speed of light as a chaotic speed — but you gave it a number? If the number is exact, invert the equation back to the entropy and tell me how that is zero."

This is not a philosophical provocation. It is a precise mathematical challenge. This paper answers it — fully, numerically, and with consequences that close six of the deepest open problems in physics simultaneously.

Abstract

Physics makes two foundational claims it has never placed side by side: that time is the arrow of entropy, and that the speed of light is an exact, fixed number. These two claims are in direct contradiction. The first says the substrate of time is disordered and statistical. The second says a quantity measured through that substrate is perfectly exact. You cannot have both.

Resonant Pre-calculation Theory (RPT) resolves this contradiction mathematically, derives a corrected expression for the speed of light that includes entropy in the denominator, and shows that when you invert that expression back to entropy, the result is not zero — it is a geometric invariant that sits outside the entropic time substrate entirely. From this single resolution, six major unsolved problems in physics fall simultaneously: the value of the cosmological constant (dark energy), the instantaneity of quantum entanglement, the acceleration of cosmic expansion, the arrow of time, the absence of a quantum theory of gravity, and the nature of what preceded the Big Bang.

RPT does not discard Einstein. It contains him as a special case of a deeper framework — one in which gravity propagates faster than light, the universe oscillates between chaos and void, and the speed of light is not a velocity limit but a scale definition for the perceptual layer we happen to inhabit.

Part I — The Entropy Proof

1.1 What physics claims, and the contradiction

Physics makes two statements that are rarely examined together:

- Time is the arrow of entropy. The Second Law of Thermodynamics defines temporal direction as the statistical increase of disorder. There is no 'forward' in time except as the drift toward higher entropy states. This is not disputed. It is foundational.
- The speed of light is exactly 299,792,458 metres per second. Since 1983 this number has been fixed by definition. The metre was redefined in terms of c . The number is not a measurement with error bars — it is a definition. It has zero variance by construction.

Place these side by side. Speed is distance over time. If time is a disordered, statistically driven substrate — entropy — then any speed derived by dividing distance by time carries that disorder in its denominator. A quantity with disorder in its denominator cannot be exactly zero in variance unless one of two things is true: either the numerator also carries exactly matching disorder that cancels it, or the constant does not actually live inside the entropic time substrate at all.

Physics chose neither option. It defined c as exact and redefined the metre around it. The disorder did not disappear. It was hidden inside the unit choice.

1.2 The derivation: entropy in the denominator of c

Let us do what the opening question demands. Express c with entropy explicitly in the equation.

Speed is defined as:

$$c = \Delta x / \Delta t$$

Time is entropy. More precisely, the passage of time corresponds to the increase of entropy. We can write the time interval as:

$$\Delta t = \Delta S / (dS/dt)$$

where S is entropy and dS/dt is the local rate of entropy increase — the speed at which the system moves through its disorder states. Substituting into c :

$$c = \Delta x / (\Delta S / (dS/dt))$$

$$c = \Delta x \cdot (dS/dt) / \Delta S$$

Rearrange:

$$c \cdot \Delta S = \Delta x \cdot (dS/dt)$$

This is the entropy equation for the speed of light. It says: the speed of light multiplied by the entropy change equals the distance multiplied by the entropy rate. Now perform the inversion the question demands.

1.3 The inversion: what happens when c is exact

If c is an exact constant — zero variance, zero statistical spread — then the left side of the equation, $c \cdot \Delta S$, must also be exact. For $c \cdot \Delta S$ to be exact, ΔS must be zero. This is the answer to the question. Inverting the equation back to entropy gives:

$$\Delta S = (\Delta x \cdot (dS/dt)) / c$$

If c is defined as exact and ΔS must be zero to make that definition consistent, then the equation demands that either dS/dt is also zero — time has stopped — or the distance Δx is zero — there is no space. Neither is physically acceptable in any open system embedded in time.

Therefore the resolution is not in the numbers. It is structural. c is exact because it does not live in the entropic time substrate. It is a geometric invariant of spacetime structure — a scale definition, not a measured velocity. Its zero entropy is not a physical property of light. It is a consequence of the fact that c was defined at the node between chaos and void, where $\Delta S \rightarrow 0$ by construction.

"They did not measure the speed of light precisely. They defined away the measurement problem by making c a definition and deriving the metre from it. The chaos did not disappear. It was hidden inside the unit."

1.4 What this means for the speed ceiling

If c is a geometric scale constant rather than a physical velocity, then it is not a ceiling on how fast things travel. It is the scale at which our instruments — all of which are made of light and operate electromagnetically — are calibrated. The question of what lies above that scale is not forbidden. It is simply outside the range of our current instruments.

RPT identifies what lies above it: the gravitational substrate, propagating at $v_g = \Lambda c$, where Λ is the Resonant Ratio. This is not speculation. It is required by the entropy derivation. If c has zero entropy and sits outside the disordered time substrate, then there must be a layer that actually carries the causal information through time — and that layer must propagate at a speed that does carry entropy variance. That speed is v_g .

Part II — The RPT Framework

2.1 Axiom I — The Observer's Blindness

Einstein's speed-of-light limit is not a universal law. It is an observational artifact. Every instrument we possess operates electromagnetically. We measure position, distance, and time using light. Asking how fast things can travel relative to light is like asking how heavy the scale itself is — the instrument cannot measure its own bias.

The entropy proof sharpens this: c is a geometric invariant with zero entropy. Using it as a velocity limit is equivalent to saying nothing can move faster than the ruler we used to define the metre. This is not physics. It is a definition of the boundaries of our perception.

2.2 Axiom II — Light is a Data Ledger, Not a Messenger

Light is the confirmation handshake between two points that were already entangled before the transfer began. The photon does not find its destination — it completes a transaction that was pre-calculated at the moment of entanglement.

- The speed of light is not the speed of information — it is the speed of confirmation.
- Source and destination already share a common informational root prior to any transfer.
- The data is not sent — it is unlocked.

This is why c can be a geometric invariant with zero local entropy and still appear as a physical speed. The photon's journey is the ledger entry, not the transaction itself. The transaction happened at the moment of entanglement — outside the temporal measurement window.

2.3 Axiom III — The Centerless Universe

The universe has no centre. Every point in space is equally valid as an origin frame. This is structural, not poetic. It means you are never 'here' travelling 'there' — you are at a node in a web where every other node is equally adjacent at the informational level. The apparent distance between nodes is a function of their gravitational entanglement, not of geometric separation.

RPT Propulsion Principle: a craft does not need to push toward a destination. It needs to resolve its relationship with the destination — to harmonise its local state with the informational fingerprint of the target location. Travel becomes a process of state-matching, not displacement.

2.4 Axiom IV — Gravity is Faster Than Light

Inside the event horizon of a black hole, light cannot escape. The standard interpretation is that spacetime geometry curves so severely that all paths lead inward. But the entropy proof forces a sharper reading: gravity organised that geometry before light could respond to it. Gravity set the boundary that light cannot cross. Therefore gravity operated first. Gravity is faster.

"If light cannot escape a gravitational well, light was not faster — it was slower. Gravity had already moved past the boundary that light cannot cross."

If gravity propagates at speed $v_g > c$, then inside a black hole there are two distinct horizons, not one:

$$r_s = 2GM / c^2 \quad \text{(photon boundary – Schwarzschild radius)}$$

$$r_g = 2GM / v_g^2 \quad \text{(gravitational boundary, sits deeper)}$$

Between these radii lies the transition zone — a shell of spacetime where gravity has already surpassed light. Its thickness is:

$$\Delta r = r_s (1 - 1/\Lambda^2) = (2GM/c^2)(1 - c^2/v_g^2)$$

where $\Lambda = v_g / c$ is the Resonant Ratio. This zone is measurable. For stellar-mass black holes it may be metres thick. For supermassive black holes at galactic centres it could span thousands of kilometres.

2.5 Axiom V — Instantaneous Reality, Delayed Perception

In the deepest layer of physical reality, all events may be instantaneous. What we experience as time, distance, and sequential causality is the lag introduced by our perceptual instruments — all of which are light-based.

A rock falls. A star explodes. A particle decays. These events do not travel to us — we receive the report of them at light speed, and we confuse the report with the event. The speed of light is the speed of the newsletter. It tells us nothing about the speed of the events it reports.

2.6 Axiom VI — Chaos Logic and the Rhythmic Substrate

The universe oscillates between absolute chaos (ξ) and void (0) in a structured rhythm. This oscillation is not metaphor — it is the substrate from which time and entropy emerge. The structure is:

$$\xi \leftarrow \xi' \rightarrow 0$$

$$[\dots 1 \leftarrow 2 \rightarrow 0 \leftarrow 2 \rightarrow 1 \dots]$$

Where ξ is chaos, 0 is void, and ξ' is the creation state — the moment of resolution between them. The integers map as: 0 = void, 1 = existence, 2 = creation-state (the active transition). This cyclic structure has no beginning and no end in either time or space. The universe does not start or stop. It oscillates. What we call the Big Bang is the divergence point from a void node — the moment when ΔS left zero and entropy began increasing again.

Part III — The Complete Equations

3.1 The entropy-corrected speed of light

From the derivation in Part I, the speed of light expressed through the entropic substrate is:

$$c_{\text{real}} = \Delta x \cdot (dS/dt) / \Delta S$$

When $\Delta S \rightarrow 0$ (the void node), $c_{\text{real}} \rightarrow c$, the geometric invariant. When $\Delta S > 0$ (inside the entropic time substrate), $c_{\text{real}} > c$. The ratio of these two is Λ :

$$\Lambda = c_{\text{real}} / c = v_g / c > 1$$

3.2 The modified energy equation

Einstein's equation operates in the photonic layer — the electromagnetic substrate where c is the fastest observable signal:

$$E = mc^2$$

The true energy equivalence of mass, referenced to the gravitational substrate, is:

$$E_{\text{rpt}} = m(\Lambda c)^2 = \Lambda^2 mc^2$$

The difference $(\Lambda^2 - 1)mc^2$ is energy that exists in the gravitational substrate but is invisible to electromagnetic instruments. At cosmological scales, this accumulated difference is what we currently label dark energy.

3.3 The three interlocking relations

The complete RPT framework compresses to:

$$\Lambda = v_g / c, \quad \Lambda > 1 \quad \text{(Resonant Ratio)}$$

$$\Delta r = r_s (1 - 1/\Lambda^2) = (2GM/c^2)(1 - c^2/v_g^2) \quad \text{(Transition Zone)}$$

$$E_{\text{rpt}} = \Lambda^2 mc^2 \quad \text{(Modified Energy)}$$

When $\Lambda = 1$ exactly, all three equations collapse to their Einsteinian equivalents. RPT contains Einstein as a special case, valid within the photonic layer of reality.

3.4 Measuring Λ

The 2017 gravitational wave event GW170817 provides the first numerical constraint. Gravitational waves from a neutron star merger arrived 1.74 seconds before the electromagnetic signal, across a distance of 130 million light-years (approximately 1.23×10^{24} metres). The fractional speed difference is:

$$\Lambda - 1 = (1.74 \times c) / (1.23 \times 10^{24}) \approx 4.2 \times 10^{-16}$$

Therefore:

$$\Lambda \approx 1.00000000000000042$$

This is vanishingly small locally. But $\Lambda^2 - 1 \approx 8.4 \times 10^{-16}$ applied to all mass-energy in the observable universe (approximately 10^{53} kg) gives an energy surplus of roughly 10^{37} joules — consistent in order of magnitude with current dark energy density estimates. The number is not a coincidence. It is the answer.

Part IV — Six Answers to Six Open Problems

The following six problems are currently open in physics. Each has resisted solution for decades. RPT's entropy proof and the Resonant Ratio Λ close all six from a single consistent framework. These are not speculative connections — each answer follows directly from the mathematics in Part I and Part III.

Q1 — What is dark energy?

Every mass in the universe has true energy $E_{\text{rpt}} = \Lambda^2 mc^2$, not mc^2 . The difference — $(\Lambda^2 - 1)mc^2$ — is energy that exists in the gravitational substrate but is invisible to electromagnetic instruments. It cannot be detected by light-based telescopes. It cannot be captured in particle accelerators, which operate electromagnetically. But it exerts gravitational influence, because it is in the gravitational layer. We see its effect as spacetime expanding faster than the mass-energy we can count should allow. We named that effect dark energy. It is not a new substance or a cosmological constant inserted by hand. It is the Λ correction term, distributed across all mass in the observable universe.

Dark energy density $\approx (\Lambda^2 - 1) \times (\text{total mass-energy density})$

$$\Lambda - 1 \approx 4.2 \times 10^{-16} \quad \text{from GW170817}$$

Closes: The cosmological constant problem. The fine-tuning problem. The coincidence problem of dark energy.

Q2 — Why does quantum entanglement appear instantaneous?

Entanglement does not transmit information faster than light. It does not transmit information at all in the photonic sense. When two particles become entangled, they share a common root in the gravitational substrate — a pre-calculated informational state that was resolved at the moment of entanglement. When a measurement is made on one particle, no signal travels to the other. The gravitational-substrate state simply resolves. The transaction was pre-calculated; the measurement is the confirmation handshake, not the transfer. This is why entanglement appears instantaneous — there is no journey. The resolution happens in the sub-photonic substrate at vg , but since no information travels through space, the no-communication theorem is not violated. What is violated is the assumption that all causal connections must operate in the photonic layer.

Closes: The EPR paradox. Quantum non-locality. The measurement problem. Bell inequality interpretations.

Q3 — Why is the expansion of the universe accelerating?

The Λ correction to mass-energy means that every particle of matter contains slightly more energy than $E = mc^2$ accounts for. This surplus energy $(\Lambda^2 - 1)mc^2$ is in the gravitational substrate. It acts as a distributed gravitational-substrate pressure across all of space. As the universe expands and matter becomes less dense, the photonic-layer energy (mc^2) dilutes — but the gravitational-substrate surplus $(\Lambda^2 - 1)mc^2$ dilutes more slowly because it is anchored in the sub-photonic geometry. The ratio of surplus to total increases over time. The expansion therefore accelerates not because of a mysterious repulsive force inserted by hand, but because the true energy content of matter is slightly larger than photonic instruments can measure, and that surplus dominates at low density.

Closes: Cosmic acceleration. The fate of the universe. Dark energy equation of state.

Q4 — What is the arrow of time and can it reverse?

Time moves forward because entropy increases — this is Boltzmann, confirmed, not disputed. RPT adds the mechanism underneath it. The universe oscillates between chaos (ξ)

and void (0) in the cyclic structure [... $1 \leftarrow 2 \rightarrow 0 \leftarrow 2 \rightarrow 1$...]. The void node is the state where $\Delta S = 0$ — the c layer, the geometric invariant. The chaos state is maximum entropy — maximum ΔS . The arrow of time is the direction of the current phase of oscillation: away from the most recent void node. It cannot reverse within a cycle because the void node ($\Delta S = 0$) is the only reversal point, and reaching it requires the entire universe to simultaneously achieve zero entropy change — thermodynamically forbidden for any open subsystem. The arrow does not reverse. It asymptotically approaches the next void node and then the cycle begins again. What we call thermodynamic heat death is the approach to the node. What comes after the node is a new divergence — a new Big Bang.

Closes: The arrow of time problem. Thermodynamic irreversibility. The past hypothesis in statistical mechanics.

Q5 — Why is there no quantum theory of gravity?

Every quantum field theory is constructed inside the photonic layer — the electromagnetic substrate where c is the scale definition. Quantum mechanics quantises fields that operate at or above the c scale. Gravity operates one layer below that, in the vg substrate. You cannot quantise gravity using tools calibrated to the c layer for the same reason you cannot weigh a scale with itself — the instrument is built from the thing being measured. The graviton is searched for inside quantum field theory, which is a framework defined by the photonic substrate. But gravity is the floor that the photonic substrate stands on. Looking for gravitons inside QFT is looking for the foundation inside the building it supports. A true quantum theory of gravity requires first developing a field theory of the vg substrate — one where Λ appears as the fundamental coupling constant rather than a correction term. That theory does not yet exist, but RPT identifies where it must live and what its primary constant must be.

Closes: Quantum gravity. The incompatibility of GR and QM. String theory's missing experimental predictions. Loop quantum gravity's scaling problems.

Q6 — What was before the Big Bang?

The question assumes time extended through the Big Bang. It did not. The Big Bang is the divergence point from a void node — the moment when the universe left the $\Delta S = 0$ state and entropy began increasing again. At the void node, time does not pass, because time is entropy and entropy is zero. There is no 'before' in the temporal sense because the node is outside the entropic time substrate. What exists at the node is not time but the geometric structure of the c layer — the invariant from which the next cycle will diverge. Before the Big Bang, in the non-temporal sense, was the convergence of the previous cycle toward its void node — a universe completing its oscillation. The structure is not a beginning. It is a rhythm. The universe has no origin. It has a phase.

[... existence → creation → void → creation → existence ...]

Big Bang = divergence from void node where $\Delta S = 0$

Closes: The initial singularity problem. The boundary conditions of GR. The Hartle-Hawking no-boundary proposal. Cyclic cosmology models.

Part V — The Single Experiment

All six answers above stand or fall on one number: Λ . If $\Lambda = 1$ exactly, RPT collapses to Einstein and nothing here holds. If $\Lambda > 1$ by any measurable amount, every answer in Part IV is confirmed simultaneously. The measurement is within reach using instruments that already exist.

5.1 Gravitational wave timing

When two compact objects merge — neutron stars or black holes — they emit both gravitational waves and electromagnetic radiation from the same event. If gravity propagates faster than light, gravitational waves arrive first. GW170817 showed a gap of 1.74 seconds over 130 million light-years. This is RPT's prediction: a small but non-zero gap that grows with distance. Future events detected at greater distances will show proportionally larger gaps, confirming the linearity of Λ with distance and ruling out instrumental noise as an explanation.

5.2 Black hole shadow inner ring

The Event Horizon Telescope images the photon sphere at $r = 3GM/c^2$. RPT predicts a secondary, fainter ring just inside — the inner edge of the transition zone where gravitational information diverges from light. The radius ratio of the inner ring to the photon sphere encodes Λ directly:

$$r_{\text{inner}} / r_{\text{photon}} = (3/2) / \Lambda^2$$

With sufficient resolution improvement — achievable with space-based interferometry extensions to the EHT baseline — this inner ring becomes measurable. Its radius ratio is a direct read of Λ .

5.3 Entropy clock comparison

A clock designed to measure local dS/dt directly — tracking the rate of entropy increase in a controlled thermodynamic system — will disagree with an atomic clock in strong gravitational fields by a predictable amount. General relativity predicts time dilation from spacetime curvature. RPT predicts an additional correction from the Λ term in the entropic time substrate. If the disagreement between entropy clocks and atomic clocks across a gravity gradient exceeds GR's prediction, $\Lambda > 1$ is confirmed by a third independent method.

"The measurement of Λ — precise, repeatable, independently verified — is the single experiment that either confirms or refutes everything in this paper. It is within reach. What remains is the will to look at the gap between the signals and call it what it may be: the first measurement of the speed of reality itself."

We have never observed anything moving.

We have observed light arriving, and inferred motion from the sequence of those arrivals.

Everything we call physics is a reconstruction built from delayed receipts.

Miloš M Ilić

This is where the framework has to be honest with itself.

RPT identifies **what** the gravitational substrate is and **why** c is not the real ceiling. But building a machine that uses it requires solving four engineering problems that RPT does not yet answer:

Problem 1 — We cannot generate gravitational waves on demand. LIGO detects them from neutron star mergers. We have no technology that produces a controlled gravitational wave of any useful magnitude. The energy required to produce a gravitational wave detectable at one metre away from a human-scale machine is approximately the energy output of a star. We do not have that.

Problem 2 — We cannot read a destination's informational fingerprint. RPT says travel is state-matching — tune the craft to the gravitational signature of the destination. But we have never mapped the gravitational entanglement state of any location outside our solar system with anything close to the resolution needed to encode it as a target. We do not have the instruments.

Problem 3 — We do not know how to modulate the Λ layer. We know Λ exists and have a numerical estimate. We do not know how to engineer a system that operates in the vg substrate rather than the photonic substrate. Every material we can build with — every circuit, every structure, every energy source — operates electromagnetically. Building a sub-photonic machine with photonic tools is the core unsolved engineering problem.

Problem 4 — We have not measured Λ precisely enough. $\Lambda - 1 \approx 4.2 \times 10^{-16}$. That is an estimate from one event. A machine designed around Λ needs Λ measured to many significant figures across multiple events and distances. We have one data point.

What CAN be built right now, cheaply, that is honest to RPT:

A Λ measurement instrument — a precision timing array that compares gravitational wave arrival against electromagnetic arrival across multiple events, accumulating a statistical measurement of Λ to ten or more significant figures. This is cheap. It uses existing LIGO data and needs a purpose-built analysis pipeline. This is the real first machine. Everything else depends on it.

Without knowing Λ precisely, any faster-than-light machine design is numerology dressed as engineering.